

# Low-overhead ML methods for beamforming/combining in RIS assisted millimeter-wave massive MIMO systems

Oumaïma Bounhar

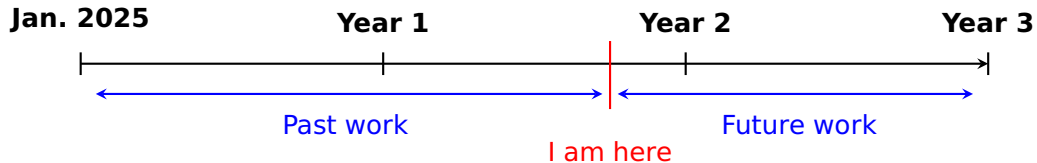
Supervised by Pr. **Ghaya Rekaya-Ben Othman**

**Télécom Paris, Institut-Mines-Télécom**, Palaiseau, France



23 septembre 2026

## Thesis Advancement & Context





## Outline

Context

System Model

State-of-the-art

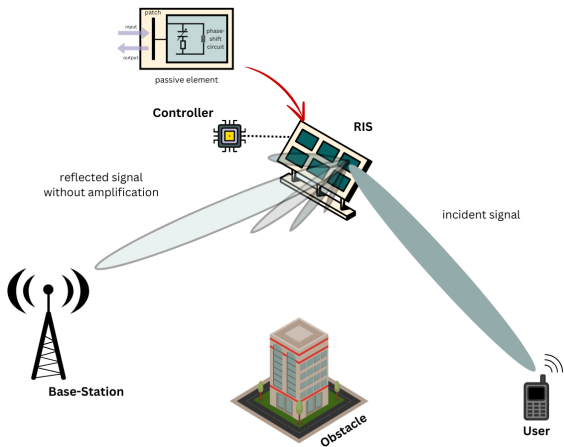
Proposed approach : Deep Reinforcement Learning

Sionna : Realistic channel generation and validation

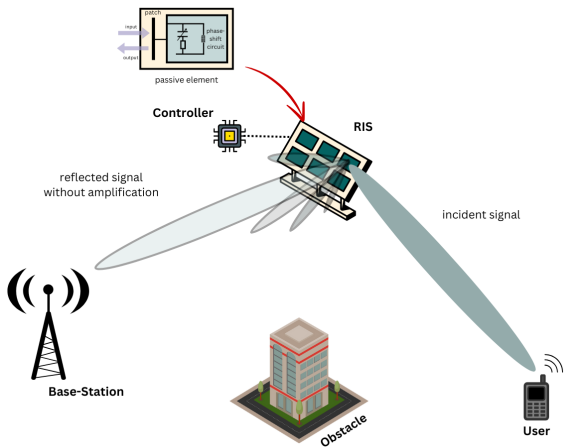
Thesis Vision Board (3 Years)

Future Perspectives

## Context

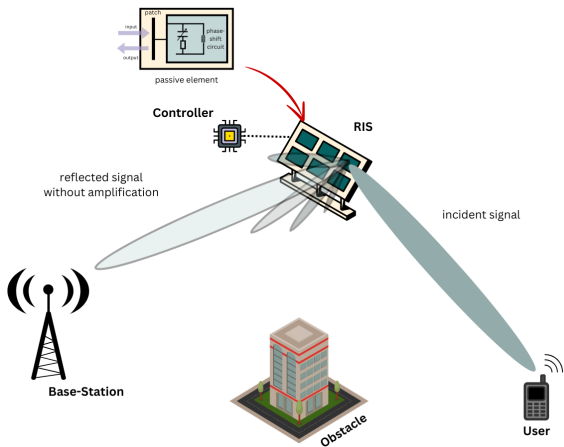


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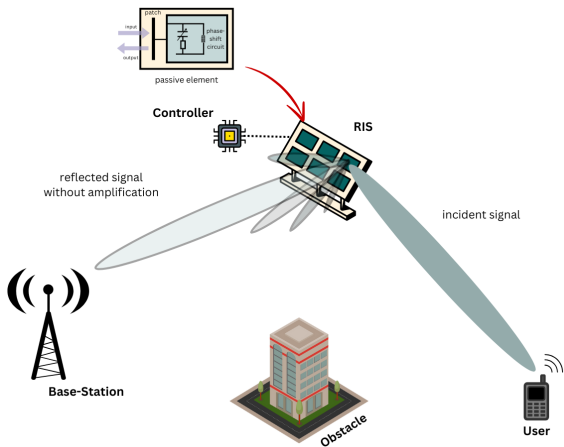
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage;

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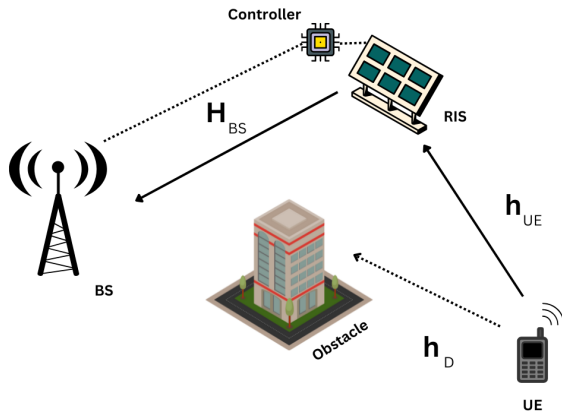
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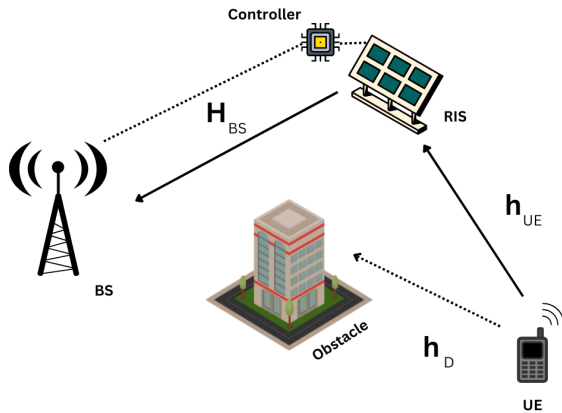
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage ;
- mmWaves suffer from fading environments ;
- RIS is a promising solution with low power consumption.

## System Model



■  $N_{UE} = 1;$

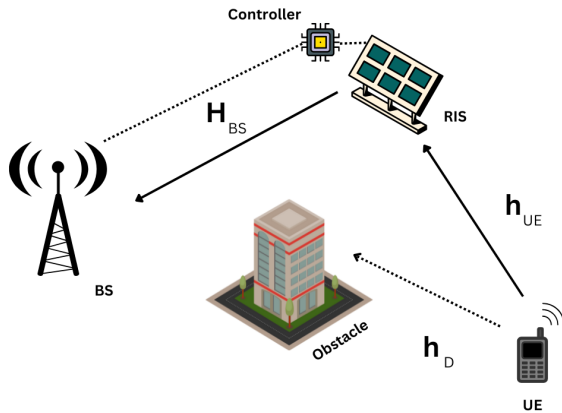
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■  $N_{UE} = 1;$

■  $N_{BS} = 64;$

## System Model

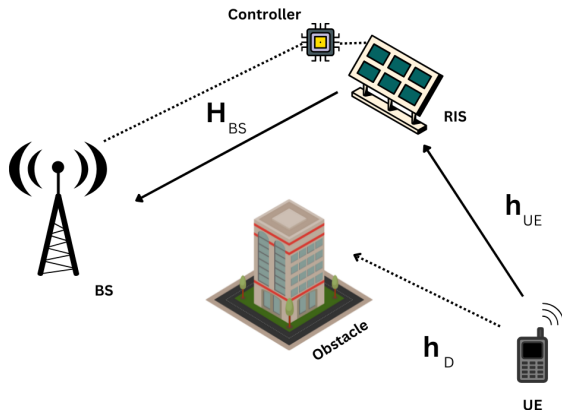


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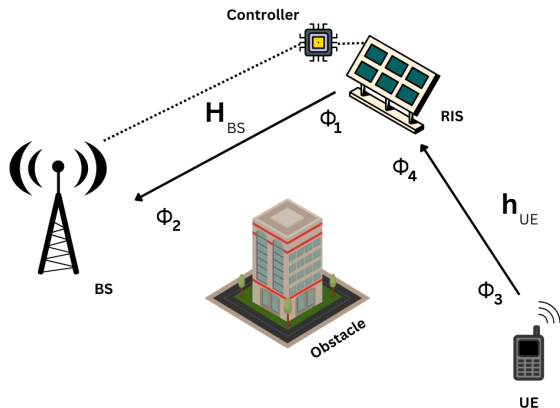
■  $N_{RIS} = 100;$

## System Model



- $N_{UE} = 1$ ;
- $N_{BS} = 64$ ;
- $N_{RIS} = 100$ ;
- The direct link is blocked.

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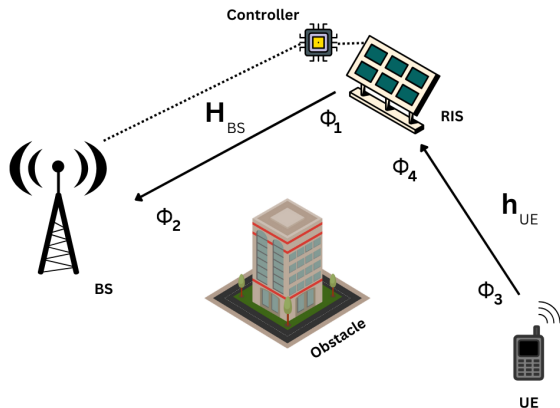
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We consider half-wave spaced ULAs :

$$\mathbf{H}_{BS} = \mathbf{a}_N(\phi_1)\mathbf{a}_M^H(\phi_2)$$

$$\mathbf{h}_{UE} = \mathbf{a}_1(\phi_3)\mathbf{a}_N^H(\phi_4)$$

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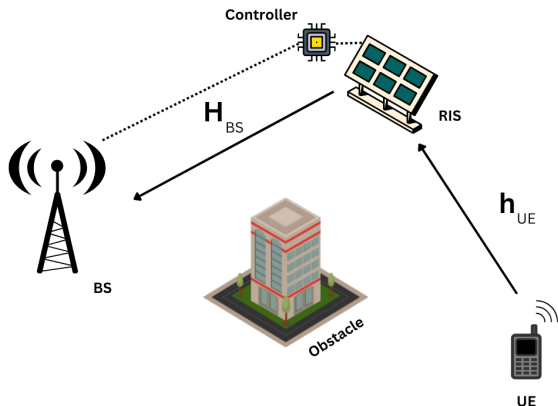
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$$\mathbf{h}_{UE} = \mathbf{a}_1(\phi_3)\mathbf{a}_N^H(\phi_4)$$

where the steering vector is :

$$\mathbf{a}_N(\Phi) = [1, e^{j\pi \sin(\Phi)}, \dots, e^{j\pi(N-1) \sin(\Phi)}]^T$$

## System Model

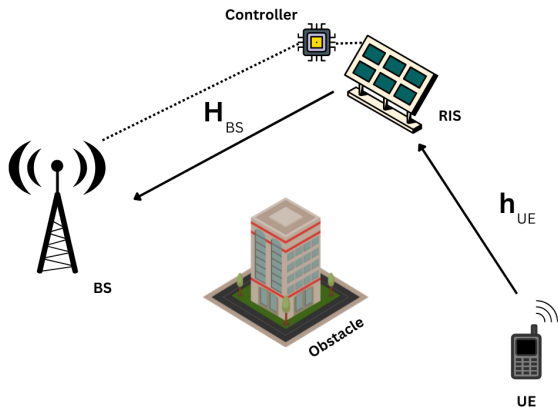


The received signal at the base station (BS) is :

$$\mathbf{y} = \sqrt{P}\mathbf{H}_{BS}\mathbf{\Theta}\mathbf{h}_{UES} + \mathbf{w}$$

where  $P$  is the transmit power of the user,  $\|s\|^2 = 1$ ,  $\mathbf{w} \sim \mathcal{CN}(0, \sigma_w^2)$  and  $\mathbf{\Theta} = \text{diag}(\mathbf{\Phi})$ , with  $\mathbf{\Phi} = [e^{j\theta_0}, \dots, e^{j\theta_{N_{RIS}-1}}]$ , where the  $\theta_i$  are the phase shifts introduced by the RIS,  $i \in [0, N_{RIS} - 1]$ .

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We write the full channel :

$$\mathbf{h}(\mathbf{\Phi}) = \mathbf{H}_{BS}\text{diag}(\mathbf{\Phi})\mathbf{h}_{UE}$$

# State-of-the-art

## Channel State Information (CSI) & Blind Beamforming

### Definition (Rate)

We want to get a higher coverage, which leads to maximizing the rate :

$$R = \max_{i \in \llbracket N_c \rrbracket} \left[ \log_2 \left( 1 + \frac{P \|\mathbf{H}_{BS} \text{diag}(\boldsymbol{\Phi}_i) \mathbf{h}_{UE}\|_2^2}{\sigma_w^2} \right) \right]$$

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### CSI

Channel estimation is challenging.

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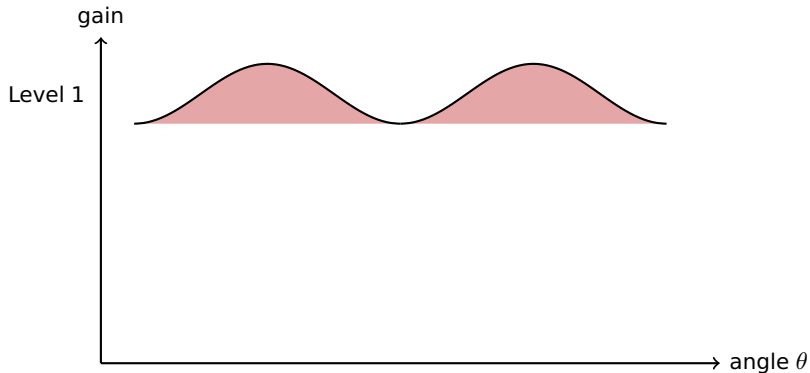
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### No CSI

"Blind" method.

# State-of-the-art

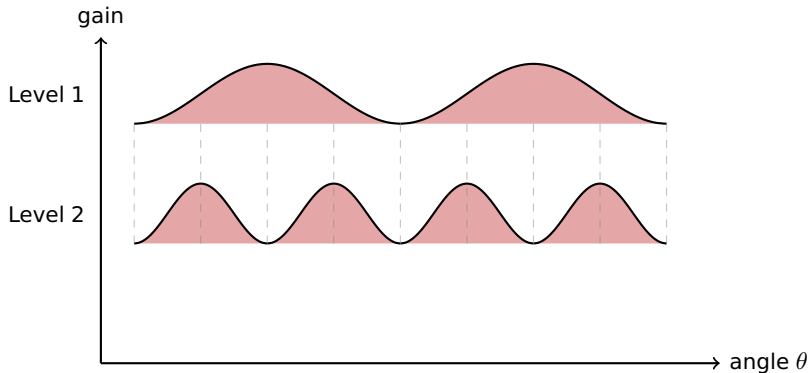
## Hierarchical codebook : What Does Each Class Represent ?



Red : successive pilot search levels over the pilot codebook  $\Psi_1^{N_p}$ .

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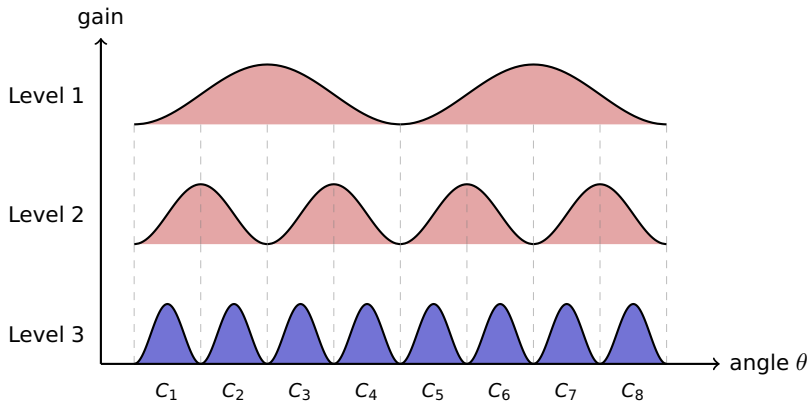
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Blue : final  $N_c = 8$  narrow DFT communication beams  $\rightarrow$  communication codebook  $\Phi_1^{N_c} \subset \Psi_1^{N_p}$ .



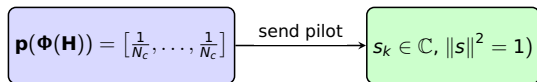
# State-of-the-art

## Bayesian Classifier : Protocol

$$\mathbf{p}(\Phi(\mathbf{H})) = \left[ \frac{1}{N_c}, \dots, \frac{1}{N_c} \right]$$

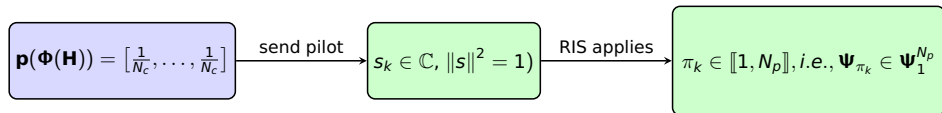
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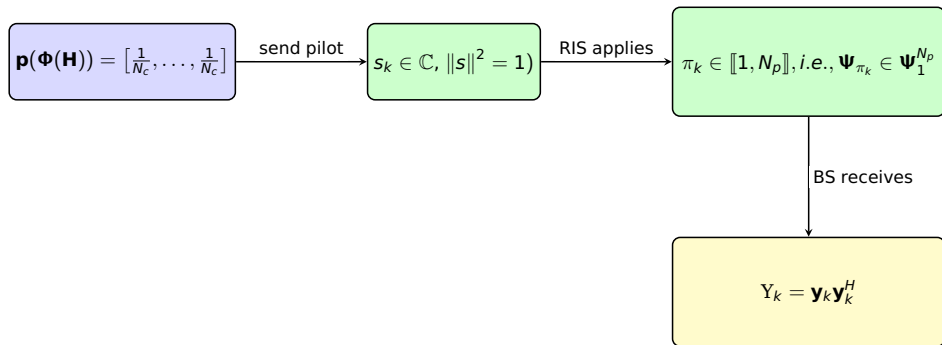
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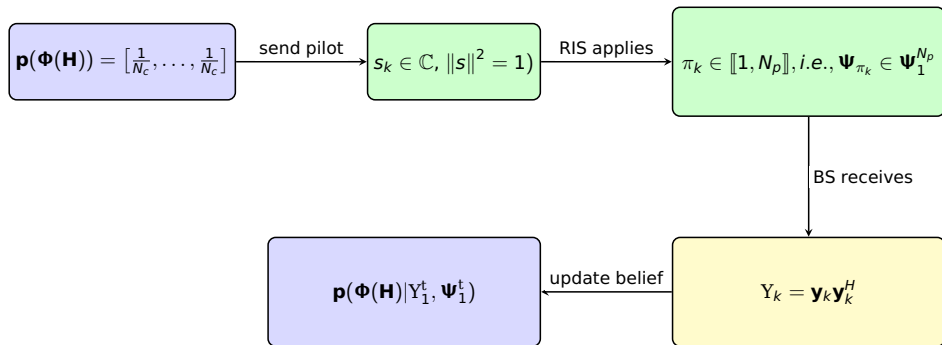
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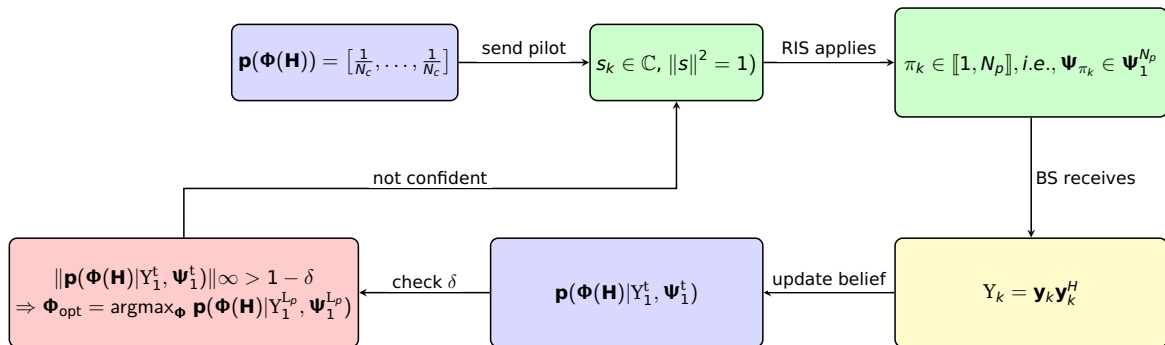
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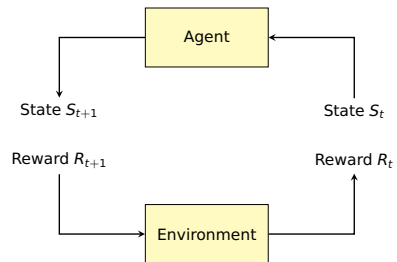
# State-of-the-art

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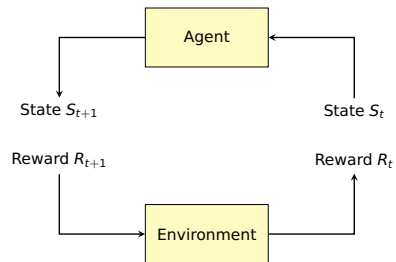
## Reinforcement Learning : State, Action, and Reward Flow



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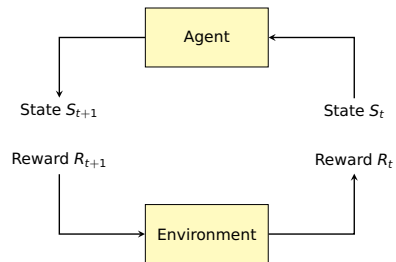
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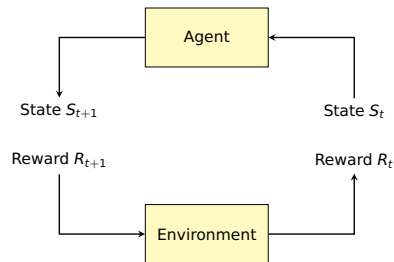
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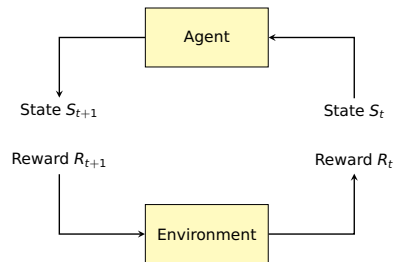
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# State-of-the-art

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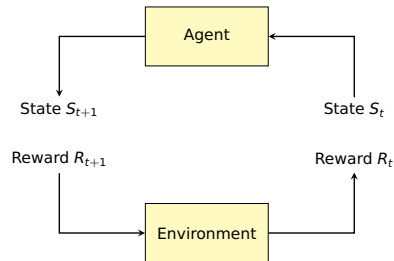
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- Reward  $\mathbf{R}_t \rightarrow -1$  until reaching  $\mathbf{S}_{\text{terminal}}$  where confidence level is  $1 - \delta$



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### Objective

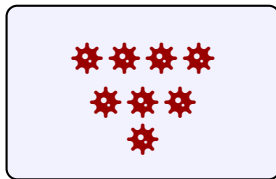
$$\Phi_{\text{opt}} = \operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})$$

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

# State-of-the-art

## A Motivating Analogy

### Reference Book (Dataset)

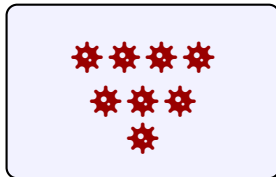


*8 known diseases*

# State-of-the-art

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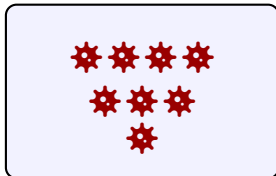


**Clinician**

# State-of-the-art

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### 14 Cheap Lab Tests

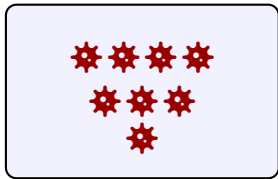


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sequentially run, one at a time

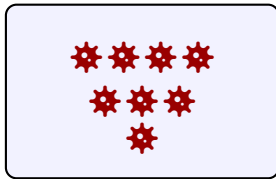


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compare results

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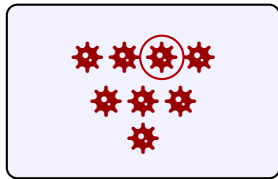


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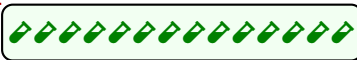


8 known diseases

Diagnosis found after  
only a few tests!

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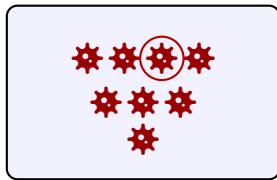


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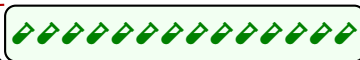


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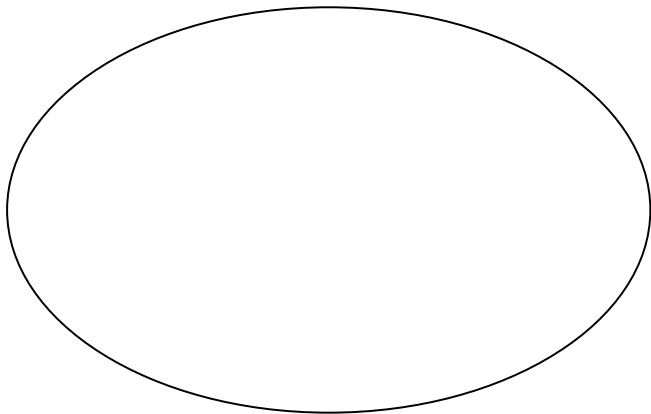
**Goal :** find the right diagnosis using as few cheap tests as possible.

# State-of-the-art

## Building the Dataset : How is a sample constructed ?

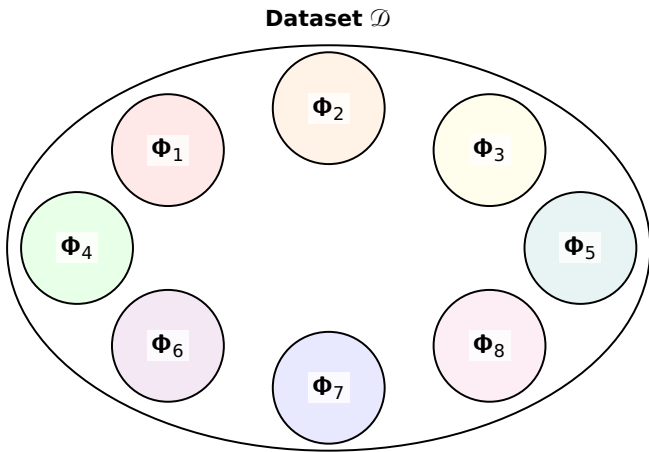
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Dataset  $\mathcal{D}$



# State-of-the-art

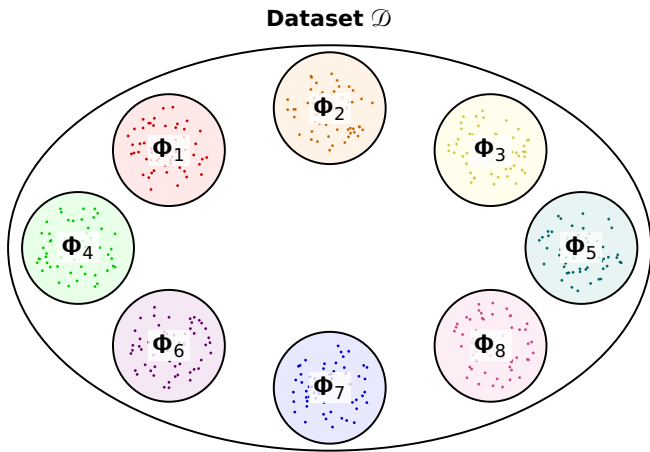
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1. Draw a random channel  $\mathbf{H}$ .
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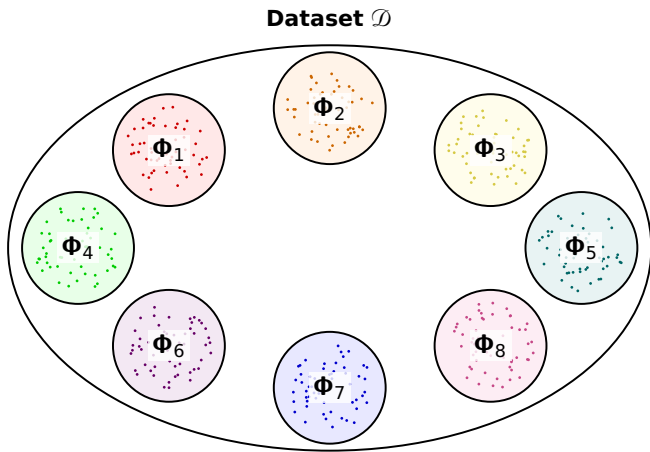
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3. Run all  $N_p=14$  pilot codewords on  $\mathbf{H}$  **with noise** (cheap tests)  $\rightarrow$  a 14-number "fingerprint" stored as a sample of class  $c^*$ .

# State-of-the-art

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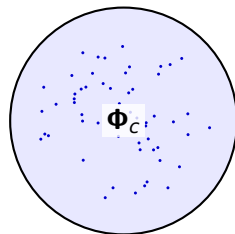
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Each class  $\Phi_c \geq 200$  such samples.

# State-of-the-art

## The Likelihood Model

This collection **is** the likelihood model.

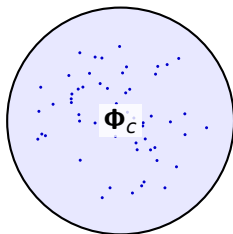


# State-of-the-art

## The Likelihood Model

This collection **is** the likelihood model.

“What do the 14 pilot readings  
tend to look like  
when the answer is  $c$ ?”



# State-of-the-art

## Stochastic shortest path problem : Policy, Objective & Q-Learning

### Policy

A policy  $\pi(S)$  determines which action  $A$  is chosen in a given state  $S$  :

$$\pi(S) = A$$

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### Q-Learning

Learn  $Q$ -function  $Q(S, A)$  from experienced transitions  $(S, A, R, S')$ , then act greedily :

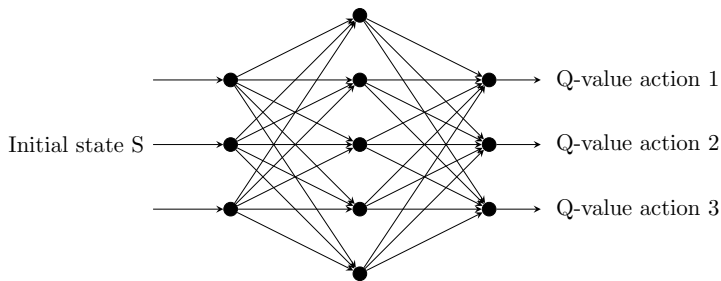
$$\underbrace{Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_{A'} Q(S', A') - Q(S, A)]}_{\text{update}} \quad \underbrace{\pi^*(S) = \arg \max_A Q^*(S, A)}_{\text{greedy policy}}$$

[2] : T. Chene, O. Bounhar, G. Rekaya, and O. Damen, "Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning," 2025.

## Proposed approach : Deep Reinforcement Learning

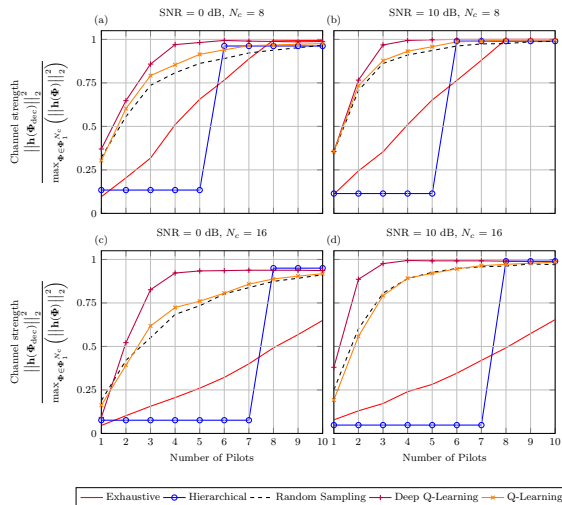
### Deep Q-Neural Network

- Replacing the Q-table with a neural network (NN) that generalizes over continuous state spaces.



# Proposed approach : Deep Reinforcement Learning

## Simulation results : Comparison to other benchmarks



[1] : O. Bounhar, T. Chene, G. Rekaya, and O. Damen, "Deep Reinforcement Learning for Blind Passive Beamforming in RIS-Assisted MIMO Systems," 2026.



## Proposed approach : Deep Reinforcement Learning

### Limitations of Deep Q-Learning

- Good for scaling size of codebook and SNR compared to QL
- Couldn't scale for multipath -> move DDQN and PPO, they behaved better than DQN for single path but still not generalizing to multipath
- cite formula of SV multipath (add details to appendix)



# Proposed approach : Deep Reinforcement Learning

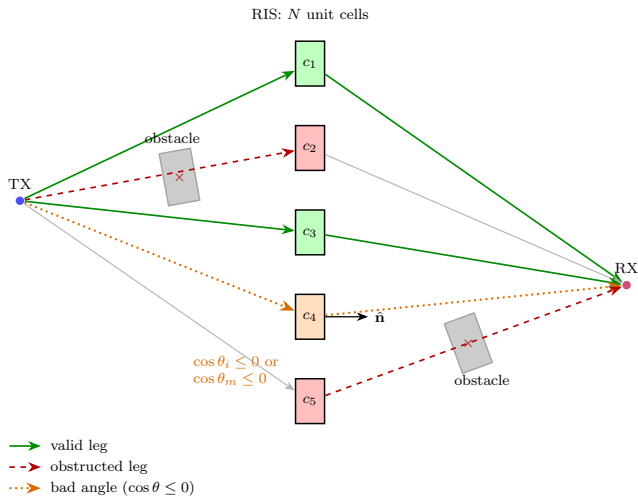
## Double Deep Q-Neural Network



# Proposed approach : Deep Reinforcement Learning Proximal Policy Optimization

# Sionna : Realistic channel generation and validation

## Ray tracing : one candidate path per unit cell



Every RIS unit cell  $c_n$ ,  $n = 1, \dots, N$ , is treated as an **independent** candidate path :

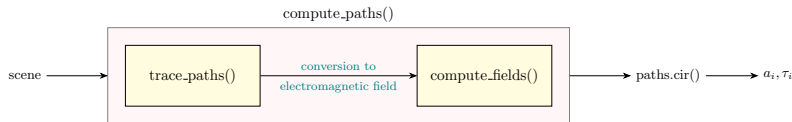
$$\text{TX} \rightarrow c_n \rightarrow \text{RX}.$$

Using the angles from the previous slide, a candidate survives only if :

- both legs are unobstructed ;
- $\cos \theta_i > 0$  and  $\cos \theta_m > 0$ .

# Sionna : Realistic channel generation and validation

## From scene geometry to channel taps



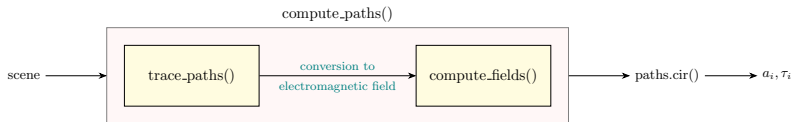
The channel impulse response is reconstructed from the  $N_{\text{valid}}$  **geometric and deterministic** per-cell taps, either kept individually resolved or coherently combined into a single tap at the mean delay :

### Individually resolved (multipath)

$$h_{\text{RIS}}(t) = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n \delta(t - \tau_n)$$

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Coherently combined (single tap)

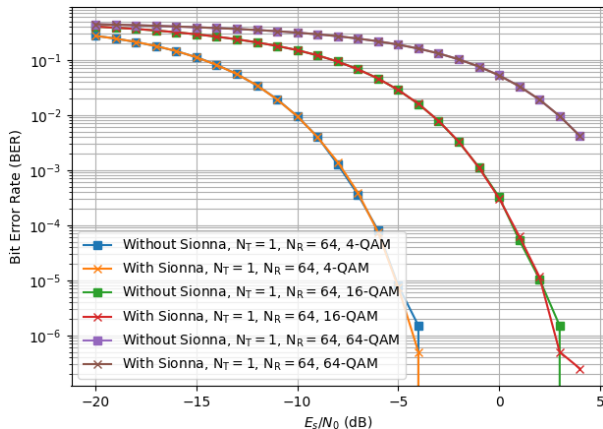
$$\bar{\tau} = \frac{1}{N_{\text{valid}}} \sum_n \tau_n$$

$$h_{\text{RIS}}(t) = a_{\text{RIS}} \delta(t - \bar{\tau})$$

$$a_{\text{RIS}} = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n e^{-j2\pi f(\tau_n - \bar{\tau})}$$

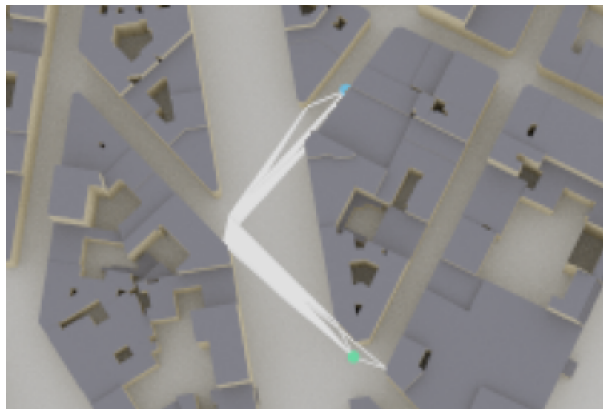
# Sionna : Realistic channel generation and validation

## BER validation : Sionna's Flat Fading vs. theoretical Rayleigh without Sionna



# Sionna : Realistic channel generation and validation

## Single-bounce per cell : RIS multipath

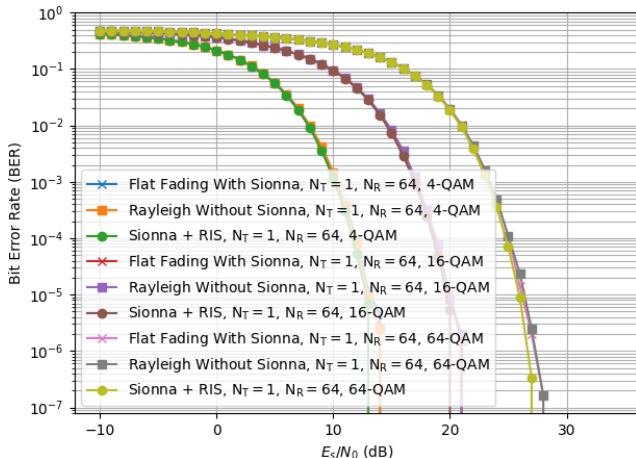


Each traced path is a single-bounce candidate  $\text{TX} \rightarrow c_n \rightarrow \text{RX}$  through one RIS unit cell  $c_n$ , obtained by ray tracing (Sionna RT) over the scene. A path is kept only if it satisfies :

- **line-of-sight** on both legs ( $\text{TX}-c_n$  and  $c_n-\text{RX}$  unobstructed) ;
- **positive illumination/re-radiation** :  $\cos \theta_i > 0$  and  $\cos \theta_m > 0$ .

# Sionna : Realistic channel generation and validation

## BER validation : RIS multipath is correctly captured



- Baselines : theoretical Rayleigh Without Sionna and Flat Fading With Sionna overlap, as before.
- Sionna + RIS confirms the RIS's multipath is correctly captured, requiring higher SNR for the same BER, across every QAM order.

## Thesis Vision Board (3 Years)

### Year 1

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- Problem formulation
- Q-Learning for RIS beamforming, 1st publication (WCNC 2025)
- ~~- Deep Q-Learning extension, 2nd publication (ICC 2026 ?)~~

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→ **Final acceptance at AIXNET [1]**

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### Year 3

- Multi-user scenario
- ~~Security aspects of~~ RIS ⌚
- Journal submission
- Thesis writing and defense

**⚠ Focus on the choice of network that allows the protocol to scale**

## Future Perspectives

### ■ *Mobile user scenario :*

- Validate the performance of our proposed approach on the generated channels with the Sionna framework.
- Handle user mobility and time-varying channels.
- Do a a real-world experiment with RIS prototypes

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- Do a a real-world experiment with RIS prototypes

### ■ *Multi-user extension :* Scale the study to multi-user scenarios.

## Scientific contributions

-  O. Bounhar, T. Chêne, G. Rekaya, and O. Damen, “Deep reinforcement learning for blind passive beamforming in ris-assisted mimo systems,” *International Conference on Interconnected AI and NETworks (AIxNET)*, 2026.
-  T. Chêne, O. Bounhar, G. Othman, Rekaya, and O. Damen, “Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning,” in *2025 IEEE Wireless Communications and Networking Conference (WCNC)*. Milan, Italy : IEEE, 2025, pp. 1–6.

# Thank you ! Questions ?

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23 septembre 2026